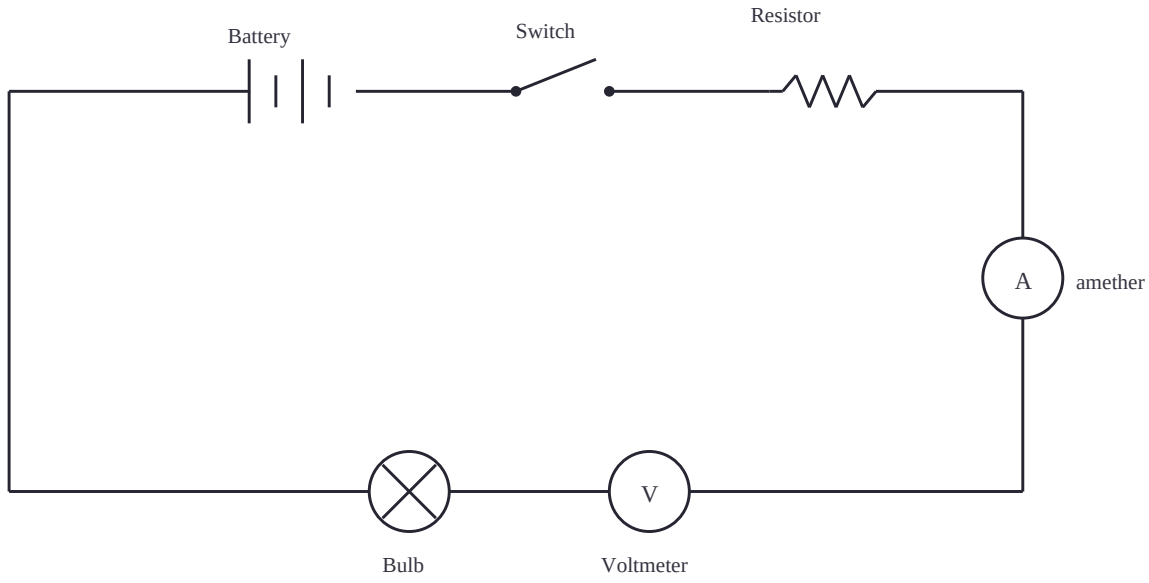
Name: Aarav Mehta Class: 10-B Subject: Combined Paper Total: 15 marks

Question 1

A simple electric circut is a closed path which allows curent to flow through it. The battery gives the potential difference which pushes the current around the loop. The switch is used to open and close the circut. When the switch is closed the path is complete and the current flows through the bulb, and when the switch is open the path is broken so no curent can flow.

> In my circut the battery, switch, resistor, bulb and the amether are all joined one after another in series in the main loop. The amether is put in series because it has to measure the curent that passes through the circut. The voltmeter is also connected in series with the bulb so that it can measure the voltage of the bulb.

If the resistance of the circut is increased then the current also increases, because a bigger resistance pushes more current through the wire.



Question 2

Technology has changed the way studnets learn today. In my opinion technology makes learning much easier and better becuase information which was very hard to find earlier is now available in a few seconds on the internet. A student who does not understand a topic in the class can watch a video at home and understand it properly. For example when I could not understand the water cycle in the class, I watched two videos on the internet and after that I understood the whole topic very well.

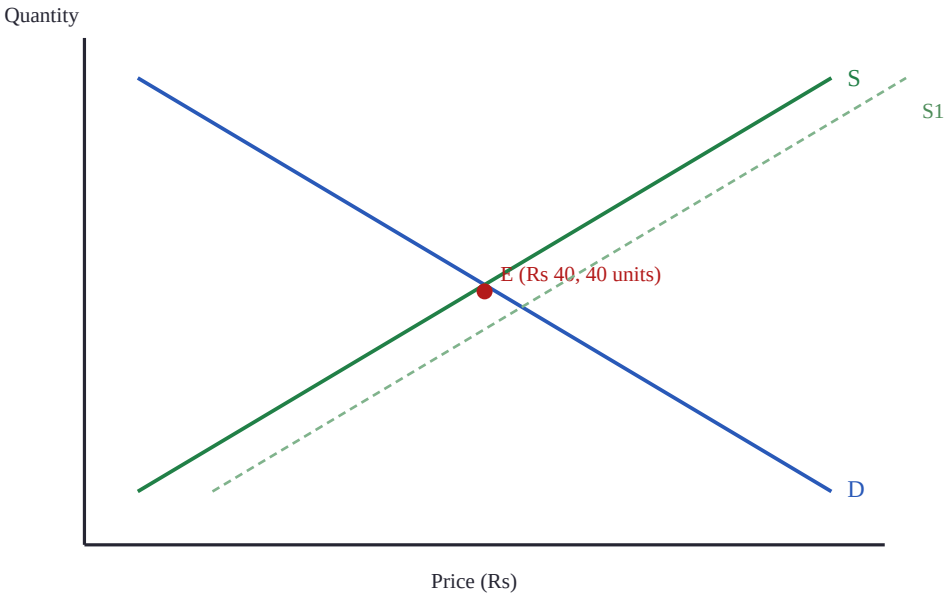
Technology also saves a lot of time becuase students do not have to search in many books in the library, they get all the informations immediately and they can spend that time on practice and revision instead which improves their knowlege and their marks in the exam also improve a lot and this is why every school should give tablets to all the students and the government should also provide free internet to every student in the country.

Question 3

The table shows the demand and the supply of the product at different prices. I have drawn the demmand curve and the suply curve on the graph below. I have taken the price on the horizontal axis and the quantity on the vertical axis. The demmand curve slopes downward because when the price is low the people buy more, and the

suply curve slopes upward because when the price is high the sellers want to sell more.

The two curves cut each other at the price of Rs 40 and the quantity of 40 units, and this point is the equilibrium of the market.

When the price is below the equilibrium price the quantity demanded becomes more than the quantity supplied, so there is a shortage in the market and the price starts rising. When the price is above the equilibrium price the quantity supplied becomes more than the quantity demanded, so there is a surplus in the market and the price starts falling.

If the cost of production increases then the producer has to spend more money on making the product, so the supply curve shifts to the right side. Because of this the new equilibrium will be at a lower price and a higher quantity than before.